Python → Data Analytics → Machine Learning

Complete Basic to Advanced Theory + Practical + Interview Preparation Guide

For Fresher Data Analyst / Data Science / ML Interviews

How to use this PDF: First understand the theory, then type and run every code example yourself, solve the practical task without looking at the solution, and finally answer the interview questions aloud.

Important: This guide follows the uploaded syllabus areas: Python, OOP, File Handling, NumPy/Pandas, Matplotlib, Machine Learning, DSA, Python + MySQL and MongoDB.

1. Python Fundamentals

Theory checklist

Syntax, comments, variables, data types, type casting, strings, booleans, operators, input/output, if/elif/else, match, while, for, range.

Practical code

```
name = input("Name: ")
age = int(input("Age: "))
if age >= 18:
    print(name, "is an adult")
else:
    print(name, "is a minor")
```

Practice tasks

1) Find whether a number is even/odd. 2) Find the largest of three numbers. 3) Print a multiplication table. 4) Count vowels in a string. 5) Reverse a string without using slicing.

Interview questions

What is Python? Why is Python dynamically typed? What is the difference between =, == and is? What are mutable and immutable objects? Explain list, tuple, set and dictionary. What is type casting? What is operator precedence? What is the difference between break, continue and pass?

2. Strings and Collections

Theory checklist

String indexing/slicing, methods, list operations, tuple packing/unpacking, set operations, dictionary keys/values/items, comprehensions.

Practical code

```
numbers = [10, 20, 10, 30, 20]
unique = list(set(numbers))
squares = [x*x for x in range(1, 6)]
print(unique)
print(squares)
```

Practice tasks

Remove duplicates from a list; find second-largest value; count frequency of words; merge two dictionaries; find common elements of two lists.

Interview questions

Why are strings immutable? List vs tuple? Set vs list? Can a list be a dictionary key? What is dictionary comprehension? What is slicing?

3. Functions, Modules and Exceptions

Theory checklist

def, parameters, arguments, return, default arguments, keyword arguments, *args, **kwargs, scope, lambda, modules, import, try/except/else/finally, raising exceptions.

Practical code

```
def calculate_average(values):
    if not values:
        return 0
    return sum(values) / len(values)

try:
    nums = [10, 20, 30]
    print(calculate_average(nums))
except Exception as e:
    print("Error:", e)
```

Practice tasks

```
Write reusable functions for factorial, prime checking, palindrome checking and frequency counting. Create a small custo
```

Interview questions

What is a function? return vs print? Local vs global variable? What are *args and **kwargs? What is lambda? What is exception handling? Why should we avoid bare except?

4. OOP in Python

Theory checklist

Class, object, __init__, self, instance/class/static methods, properties, inheritance, polymorphism, encapsulation, magic methods, inner classes.

Practical code

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def result(self):
        return "Pass" if self.marks >= 40 else "Fail"

s = Student("Mahesh", 75)
print(s.name, s.result())
```

Practice tasks

Create BankAccount, Employee and Product classes. Add methods for deposit/withdraw, salary calculation and discount calculation.

Interview questions

What is OOP? Class vs object? Purpose of __init__? Why self? Explain inheritance and polymorphism with examples. What is encapsulation? What are magic methods?

5. File Handling

Theory checklist

open(), read(), readline(), readlines(), write(), append, with statement, CSV/JSON basics.

Practical code

```
with open("students.txt", "w") as f:
    f.write("Amit,80\n")
    f.write("Riya,90\n")

with open("students.txt", "r") as f:
    print(f.read())
```

Practice tasks

Read a CSV file, calculate average marks, create a cleaned output file, and handle missing/invalid records.

Interview questions

Why use with open()? Difference between r, w and a? How do you safely handle file errors? How would you process a large file?

6. NumPy

Theory checklist

ndarray, shape, dimensions, dtype, indexing, slicing, reshape, aggregation, broadcasting, vectorization.

Practical code

```
import numpy as np

a = np.array([10, 20, 30, 40])
print(a.mean())
print(a[a > 20])
print(a.reshape(2, 2))
```

Practice tasks

Create a 3×3 array, calculate row/column means, replace values above a threshold, and normalize a numeric array.

Interview questions

Why NumPy instead of Python lists for numerical work? What is vectorization? What is broadcasting? Difference between shape and size?

7. Pandas and Data Analysis

Theory checklist

Series, DataFrame, read_csv, head/tail/info/describe, loc/iloc, filtering, missing values, duplicates, sorting, groupby, aggregation, merge, concat, apply.

Practical code

```
import pandas as pd

df = pd.DataFrame({
    "city": ["Pune", "Pune", "Mumbai"],
    "sales": [100, 150, 200]
})

print(df.groupby("city")["sales"].sum())
print(df[df["sales"] > 120])
```

Practice tasks

Load a CSV; inspect data types; find missing values; remove duplicates; group sales by category; merge customer and transaction tables; produce a final summary table.

Interview questions

Series vs DataFrame? loc vs iloc? groupby? merge vs concat? How do you handle missing values? How do you detect duplicates? apply vs vectorized operations?

8. Matplotlib and Visualization

Theory checklist

Line, bar, scatter, histogram, pie, labels, title, legend, grid, subplots.

Practical code

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr"]
sales = [120, 150, 130, 180]

plt.plot(months, sales, marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Practice tasks

Create charts for monthly sales, category distribution, age distribution and feature relationships. Explain why each chart is appropriate.

Interview questions

When would you use bar vs line vs histogram vs scatter plot? What is a subplot? How do you label axes and explain a visualization?

9. Statistics for ML

Theory checklist

Mean, median, mode, range, variance, standard deviation, percentile, distributions, normal distribution, outliers, correlation.

Practical tasks

Calculate summary statistics for a dataset, detect possible outliers, compare distributions and explain whether mean or median is more suitable.

Interview questions

Mean vs median? What does standard deviation tell us? What is a percentile? What is correlation? Does correlation imply causation? What is an outlier?

10. Machine Learning Fundamentals

Theory checklist

Supervised vs unsupervised learning, features, target, train/test split, preprocessing, scaling, categorical encoding, leakage, overfitting/underfitting.

Practical workflow

Load data → audit → clean → split X/y → train/test split → preprocess → train model → predict → evaluate → tune → save pipeline → test on unseen data.

Interview questions

What is ML? Supervised vs unsupervised? Why split data? What is data leakage? Why scale features? What is overfitting? How can it be reduced? What is a pipeline?

11. Regression

Theory checklist

Linear regression, multiple regression, polynomial regression, MAE, MSE, RMSE, R^2.

Practical tasks

Build a house-price regression model, compare MAE/RMSE/R^2, inspect residual errors, and explain model limitations.

Interview questions

What is linear regression? What does R^2 mean? MAE vs RMSE? Why polynomial regression? What assumptions are important?

12. Classification

Theory checklist

Logistic regression, KNN, Decision Tree, probability, threshold, confusion matrix, precision, recall, F1, ROC-AUC.

Practical tasks

Build a binary classifier, create a confusion matrix, calculate precision/recall/F1 and explain the business impact of false positives and false negatives.

Interview questions

Why is logistic regression used for classification? Precision vs recall? What is F1? What is a confusion matrix? What is ROC-AUC? How does KNN work? Decision Tree advantages and overfitting?

13. Clustering and Ensemble Methods

Theory checklist

K-Means, hierarchical clustering, bootstrap aggregation/bagging, cluster interpretation.

Practical tasks

Cluster customers using numerical features, select a reasonable number of clusters, visualize them and explain business interpretation.

Interview questions

How does K-Means work? What is a centroid? What is hierarchical clustering? What is bagging? Why can trees overfit?

14. Model Selection and Tuning

Theory checklist

Cross-validation, Grid Search, hyperparameters, validation data, model comparison.

Practical code

```
from sklearn.model_selection import GridSearchCV

# Example idea:
# grid = GridSearchCV(model, {"max_depth": [3, 5, 7]}, cv=5)
# grid.fit(X_train, y_train)
# print(grid.best_params_)
```

Interview questions

What is cross-validation? Why use it? Parameter vs hyperparameter? What is GridSearchCV? Why should test data remain untouched during tuning?

15. Python DSA

Theory checklist

Arrays/lists, stacks, queues, linked lists, hash tables, trees, BST, AVL concept, graphs, linear/binary search, bubble/selection/insertion/quick/merge/counting/radix sort.

Practical tasks

Implement stack and queue; reverse a linked list; binary search a sorted list; find duplicates; solve two-sum; implement at least two sorting algorithms.

Interview questions

Stack vs queue? Array vs linked list? What is a hash table? Linear vs binary search? What is recursion? What is a binary tree? BST? What is time complexity?

16. Python + MySQL

Theory checklist

Database, table, CRUD, SELECT, WHERE, ORDER BY, LIMIT, UPDATE, DELETE, JOIN, Python database connection.

Practical tasks

Create a retail database, load transactions, query top customers, monthly sales, category totals, and join customer/order tables from Python.

Interview questions

What is a primary key? What is a JOIN? WHERE vs HAVING? INNER vs LEFT JOIN? Why parameterized queries? How does Python connect to MySQL?

17. MongoDB

Theory checklist

Database, collection, document, insert, find, query, sort, update, delete, limit.

Practical tasks

Create a transaction collection, insert documents, filter by amount/category, sort by timestamp, update a document and delete invalid records.

Interview questions

SQL vs MongoDB? Database vs collection? Document vs row? Why use MongoDB? What is a query? How do update and delete operations work?

18. Real Interview Practical Round

Common scenarios

A CSV has missing values—what do you do? A model has 99% accuracy but poor fraud detection—why? A query is slow—how would you investigate? A dataset contains duplicate transactions—how do you detect them? Your model works locally but fails in production—what could cause it?

Project explanation framework

Problem → Dataset → Data cleaning → EDA → Features → Model → Evaluation → Result → Limitations → Deployment → Future improvements.

Fresher interview questions

Tell me about yourself. Explain your strongest project. Why did you choose this algorithm? What challenges did you face? How did you validate your result? What would you improve with more time? How would your project be used in a real company?

19. Final Practice Checklist

Daily routine

30–45 min theory + 60 min coding + 30 min interview questions + 15 min revision.

Minimum hands-on projects

1 Python automation project, 1 EDA/Data Analytics project, 1 SQL project, 1 ML classification project, 1 ML regression project, and 1 end-to-end fraud detection project.

Interview readiness

You should be able to explain every project without reading, write basic Python/SQL live, interpret model metrics, clean a dataset, and explain why you selected a method.